

Drive Blasting

09/05/2019

Primer Explosive I

- *Blast holes are charged with continuous column of explosive.
- *Blasting effectiveness is greatly influenced by location of primer in explosive column.
- *Stemming material & its length is also matters for blasting operation.
- *Primer placed at collar of explosive column – direct initiation
- *In direct initiation, blasting energy utilization is poor, hence less effective blasting.
- *In inverse initiation, blasting energy utilization is better.

Primer Explosive II

*With increasing depth of hole, effectiveness of blasting goes up.

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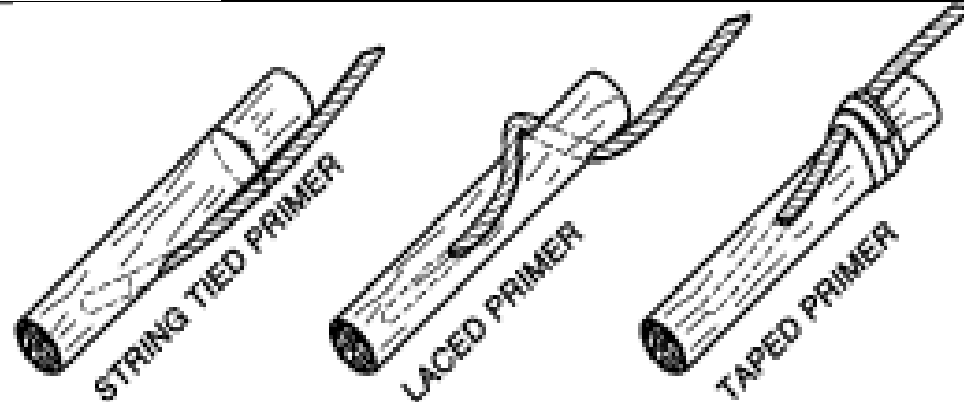
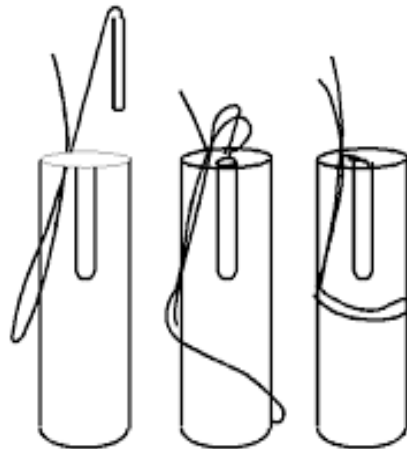
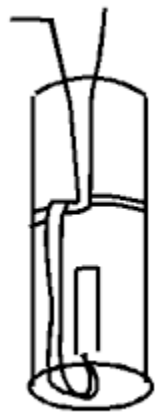
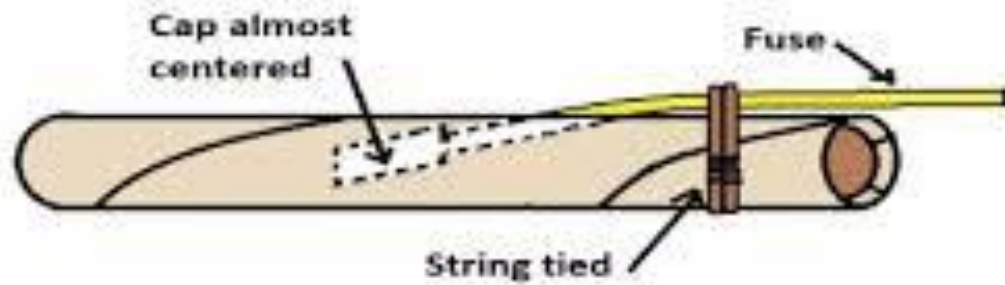
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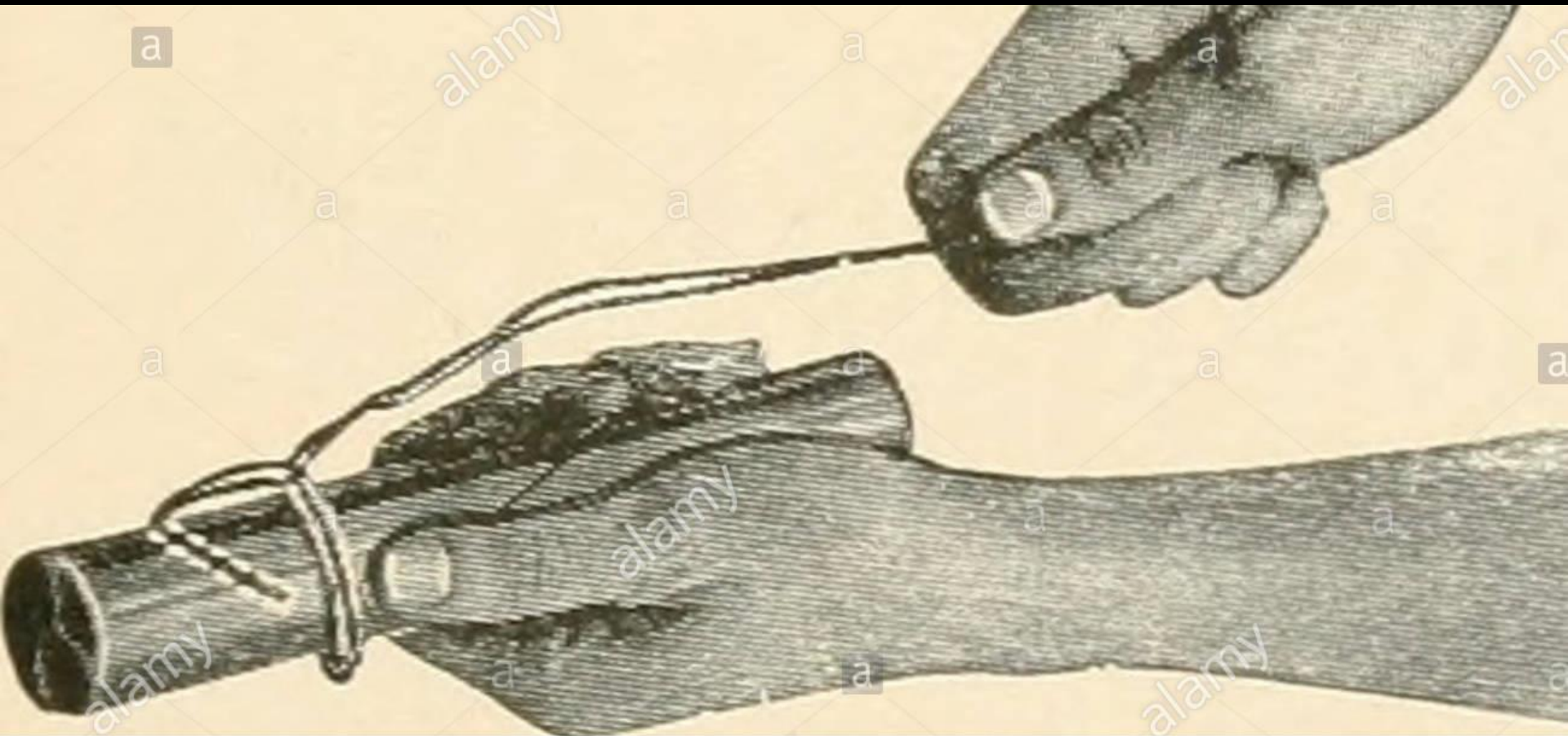
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Primer Explosive



Primer Explosive



Stemming I

- *After charging explosive in drill hole, empty portion of hole is plugged with stemming material.
- *Stemming improves blasting efficiency by increasing confinement.
- *Stemming material is made of clay : sand : water at 1:4:1.
- *Water stemming (polyethylene tube dilled with water) contributes in absorbing gas & dust from explotion.

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Depth Of Hole I

*Depth of hole is $0.5\sqrt{S}$ for angular cut hole.

*For parallel cut hole, depth is $0.75\sqrt{S}$

*S= cross section of drive.

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Charge Density I

- *Charge density is quantity of explosive per meter length of hole in cut hole area.
- *Charge is function of diameter of empty hole, its distance from charge hole & explosive density.
- *Distance between cut hole & volume of rock likely to be blasted are very small at cut hole area.
- *High charge density at cut hole area, compress rock in to plastic state, thus failing to break rock.

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Charge Density II

- *This condition reduces requirement of charge concentration per meter hole length very low.
- *Explosive of low density should be used for blasting.
- *While high density explosive loose configuration for explosive charging should be followed.

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Charge Density III

- *High density Explosive should not be stamped or pressed hard in blast hole & spacers are to be used in blast hole.
- *High density explosive are of explosive at $1.2 - 1.7$ gm/cc density.
- *Cut area after blasting creates a sufficient void / cavity.
- *Rest of holes, which are drilled at an increasing burden & spacing, yielding sufficient rock, should be charged with explosive by tapping.

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Charge Density IV

- *At line holes (peripheral holes), spacing between holes should be reduced to obtain smooth profile with minimum breakage.
- *For hole diameter in range of 32-40 mm, explosive VOD is in range of 2000-3000 meter per second.
- *In larger diameter hole, VOD of explosive should be higher.
- *Higher cross section area of drive reduces explosive consumption for cubic meter of excavation.

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Smooth Blasting I

*Smooth blasting has following effects

*Reduce cost in supporting.

*Reduce mucking.

*Reduce Dilution.

*Reduce generation of cracks & roof fall.

*Smooth wall reduces fractional resistance to ventilation air, water in hydraulic power station..

*Number of peripheral (side holes) are increased & space between them are reduced.

*Spicing & burden of peripheral holes are 0.5 to 0.6m

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Smooth Blasting II

- *Peripheral holes should be located as close to drive wall as drilling machine permits.
- *In smooth blasting, outer profile drilling must be increased to 1.5 times normal drilling.
- *Drilling accuracy of hand held machine is poor, resulting non smooth wall.
- *Drill jumbo can provide accurate drilling.
- *For charging holes for smooth blasting, special explosives are available in some countries.
- *This explosives are supplied in small diameter rigid pipes.

Smooth Blasting III

- *Pre splitting is used to obtain smooth side wall.
- *Pre splitting consisting of drilling peripheral hole with close spacing in single line or plane.
- *Explosive is charged in hole at half explosive & half wood / bamboo ?????.
- *Wood or bamboo remains towards out side wall.
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Charging & Blasting Procedure I

- *Before charging primer is prepared after collecting quantity of explosive required.
- *Blasting & explosive charging tools are collected.
- *List of tools are provided below.
- *Scraper for scrapping out loose material from blast hole.

Charging & Blasting Procedure IA

- *Pickers.

- *Stemming Road.

- *Circuit tester. Ohm meter.

- *Connecting wire.

- *Blasting cable.

Charging & Blasting Procedure II

- *Warning Display board.

- *Hooter.

- *Exploder (exploding device).

- *Shot holes are checked for correct numbers, position of holes, length of holes etc.

- *Shot holes are cleaned by blowing them with compressed air or water.

- *This flushes away cutting from blast holes.

- *Tamp (ram) explosive cartridges as required based on location of holes in respect to cut holes.

Charging & Blasting Procedure III

- *Make tight & neat connection of lead wire of detonators.
- *Joints of connection should not come to touch ground or any other metallic item at place of charging.
- *Charging equipment is to be properly earthed, if charging is done by pneumatic machine.
- *Before blasting, lay blasting cable properly & test circuit for correctness.
- *Post guard at appropriate location.
- *Display warning display board.
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Charging & Blasting Procedure IV

- *Before blasting ring siren or hooter to be used.
- *Take shelter at two right angle from blasting face, where ever practicable.
- *Reverse ventilation current to cut this exhaust, if required.
- *All precautions are to be taken before turning key of exploder.
- *After blasting allow sufficient time before re-approaching face.

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Charging & Blasting Procedure V

*Check for misfire, if any.

*In case there is any misfire, standard procedure to be followed for misfire treatment.

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Use Of ANFO In Drive I

- *Holes less than 40mm dia charged with ANFO may not give proper blasting result.
- *When hole diameter is more than 40mm, ANFO explosive is cheaper & productive.
- *ANFO is charged pneumatically using ejector type loader & anti static detonator to avoid risk of static charge that is produced during pneumatic charging.
- *Use of plastic ignitor cable (non electric detonator) & cord can be alternative to anti static detonator.

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Muck Disposal From Drive I

- *After blasting once fume is cleared & face is inspected by supervisor & blaster.
- *If face is free from misfire, water spraying is done at face to suppresses dust.
- *In case misfire is found at face, face has to be kept fenced, till misfire is retreaded and face declared safe.
- *Face is subjected to loose dressing (scaled) & is made ready for mucking.
- *Selection of mucking & transport unit differ from mine to mine.

Muck Disposal From Drive II

- *At some places, same mine can have different mucking equipment.
- *Small mines have rocker shovel for mucking & transportation by locomotive.
- *In medium & large production mine loading is done by LHD (load haul dump) & transportation is done by LPDT (low profile dump truck), conveyer, rail.
- *During muck disposal from drive, proper matching of equipment is important to avoid delay & weighting time.

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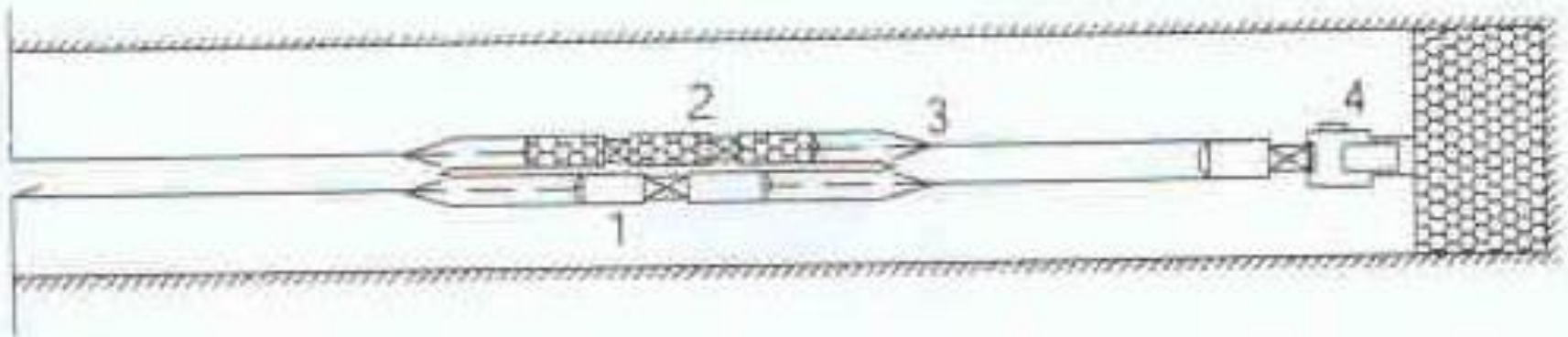
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Muck Disposal From Drive III

- *At time of purchasing equipment, decision has to be taken judiciously.
- *Layout of road way has to be made carefully, so that mucking is swift, efficient & consumes least time.
- *While driving blind heading drive, muck disposal arrangement with trackless equipment requires extra space & room head at regular intervals.
- *These space is required for transfer & unloading point.
- *Present day LHD bucket comes with pusher plate, so that requirement of additional space for unloading is reduce.

Muck Disposal I

*Rock

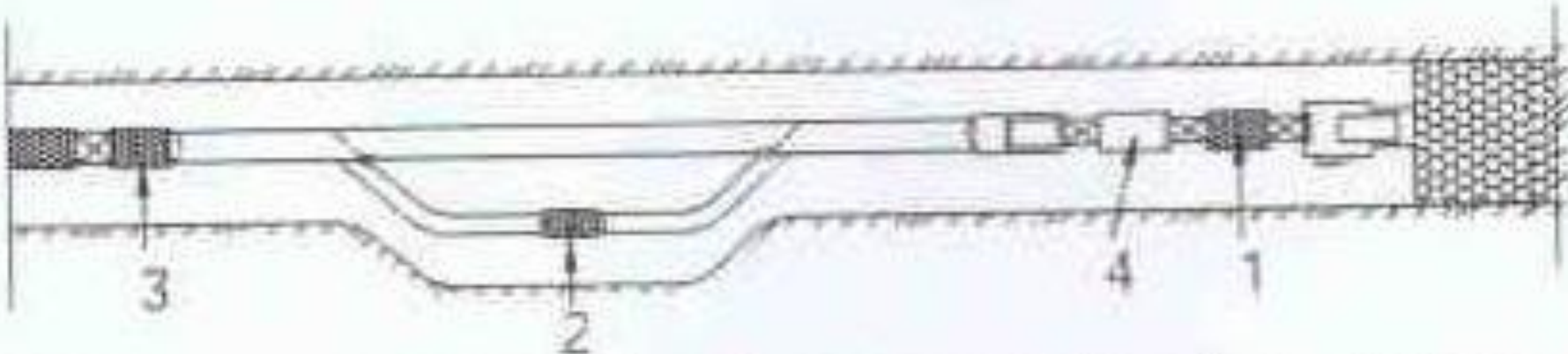


(a) Singletrack system with closed superimposed parting

1 – Empty car	2 – Loaded car	3 – Superimposed parting	4 – Loader
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Muck Disposal I

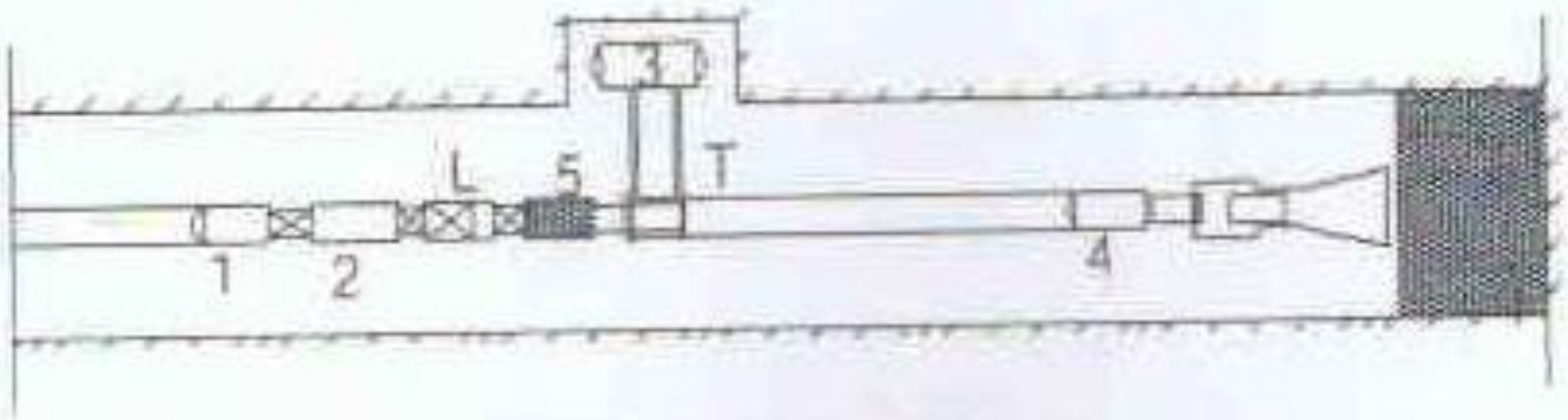
*Rock



(b) Single track system with bye-pass system, 1-4 mine cars

Muck Disposal I

*Rock



(c) Single track system using traverser

L – locomotive; T – Traverser; 1–5 single cars

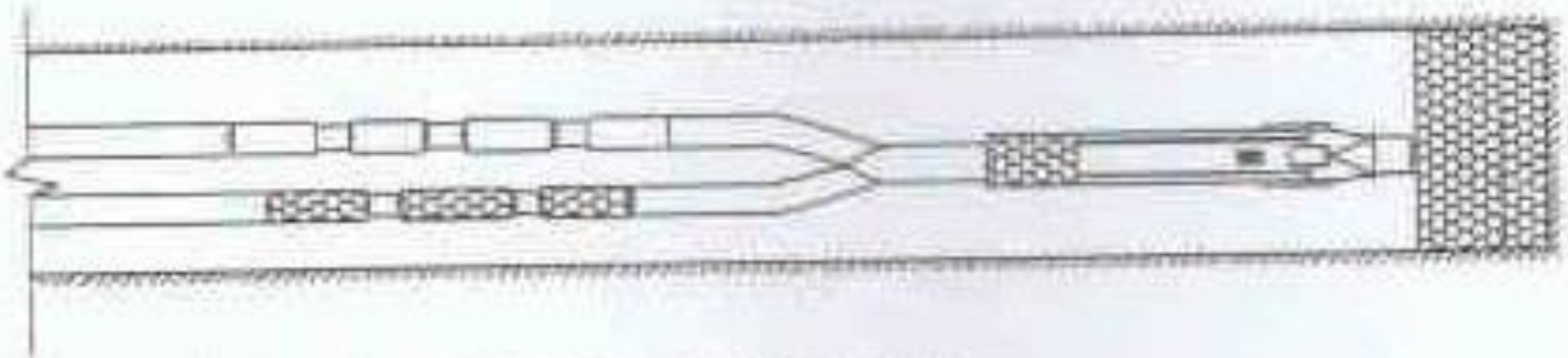
Turn Table I



Turn Table I

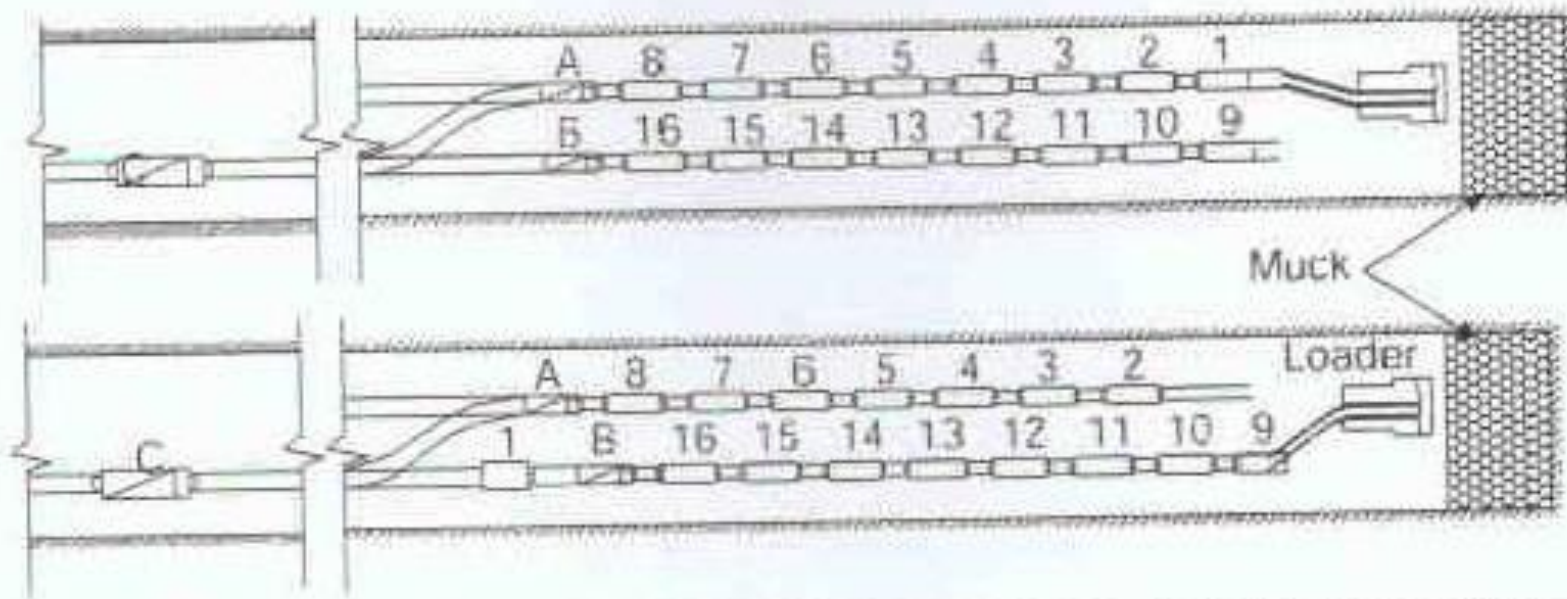


Muck Disposal I



(d) Double track using portable turnout

Muck Disposal I



(e) Double track using a pair of shunting locomotives. 1–8 mine cars (first train); 9–16 mine cars (second train); A & B – Shunting locomotives; C – Main locomotive.

Rocker Shovel Tire Mounted I



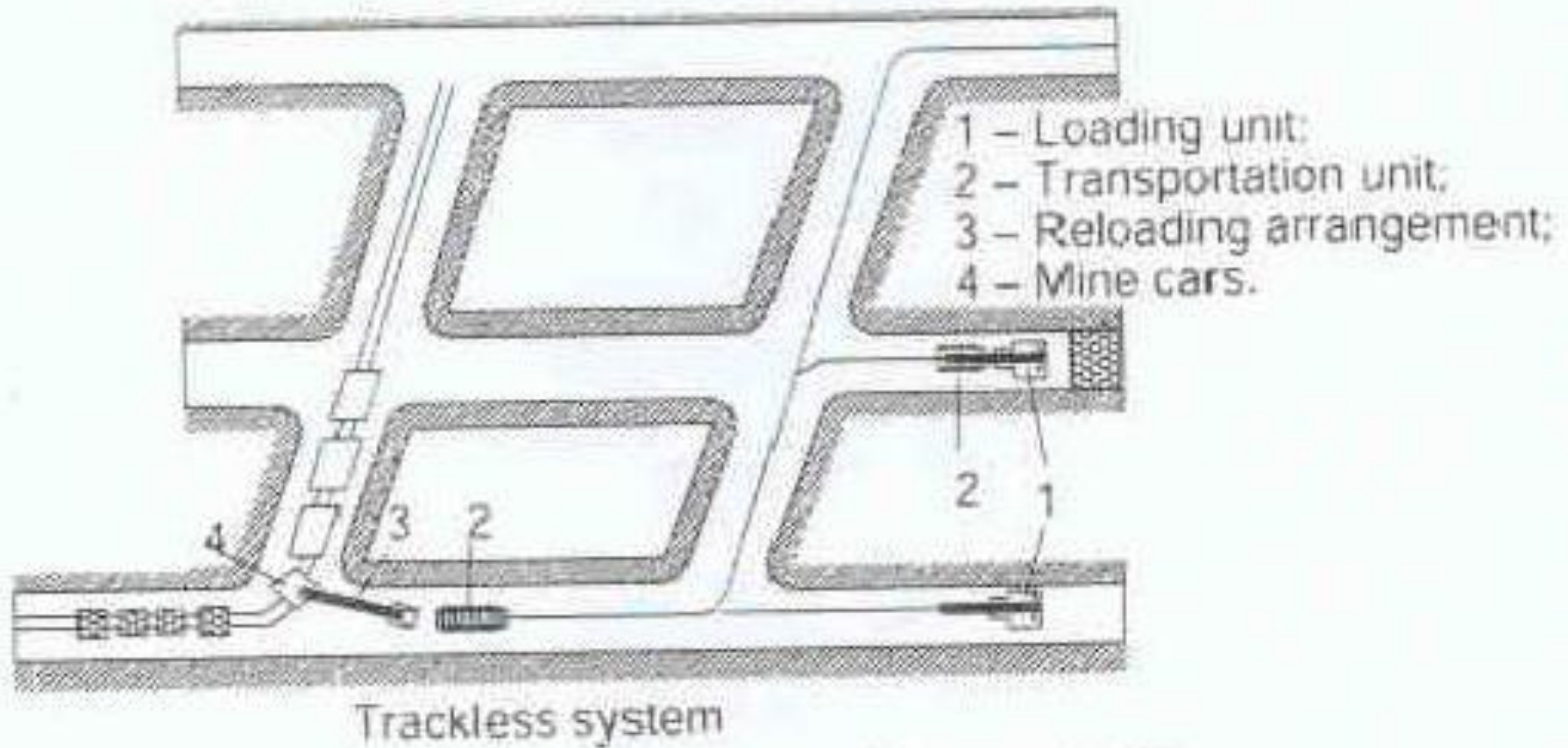
Rocker Shovel track Mounted & Mine Car

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Track Less Loader With Mine Car

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(f) Muck handling by trackless loading and transportation units from single or multiple headings.

Diesel Load Haul Dump (LHD)

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Diesel Low Profile Dump truck

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Loading Arrangement

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Ventilation By General Air Flow I

- *This system can be practiced if length of drive from opening is small.
- *Air is directed towards fan by use of some devices like baffles, screen, duct, door or other contravenes.
- *This system proves not to be very efficient & ventilation at blind end remains sluggish.

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Ventilation by Forcing Fan I

- *Auxiliary fans are used to ventilate blind drive .
- *Auxiliary fan can be contrarotating to achieve high pressure.
- *High pressure is required to send ventilation air for long distance.
- *Fresh air is discharged at work face through ventilation duct.
- *Fresh air at high pressure gets to work face & foul air is pushed out from there through drive as return air.
- *To achieve effective ventilation at work face, fan duct should not be located more than 10m from face.
- *

Ventilation by Forcing Fan II

- *Auxiliary fan is located at main drive & at least 15m from junction, to ward intake side
- *Ventilation duct, when kept closed to face, ventilation duct has to be removed before every blasting or duct likely to get damage.
- *In case of long drive, foul air has to travel along drive at low velocity.
- *This results, long face clearance time.
- *An exhaust system in side drive solves this problem by taking foul air in duct.

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Ventilation by Forcing Fan II

*Along with exhaust fan, a forcing fan with duct in drive provides best possible ventilation system.

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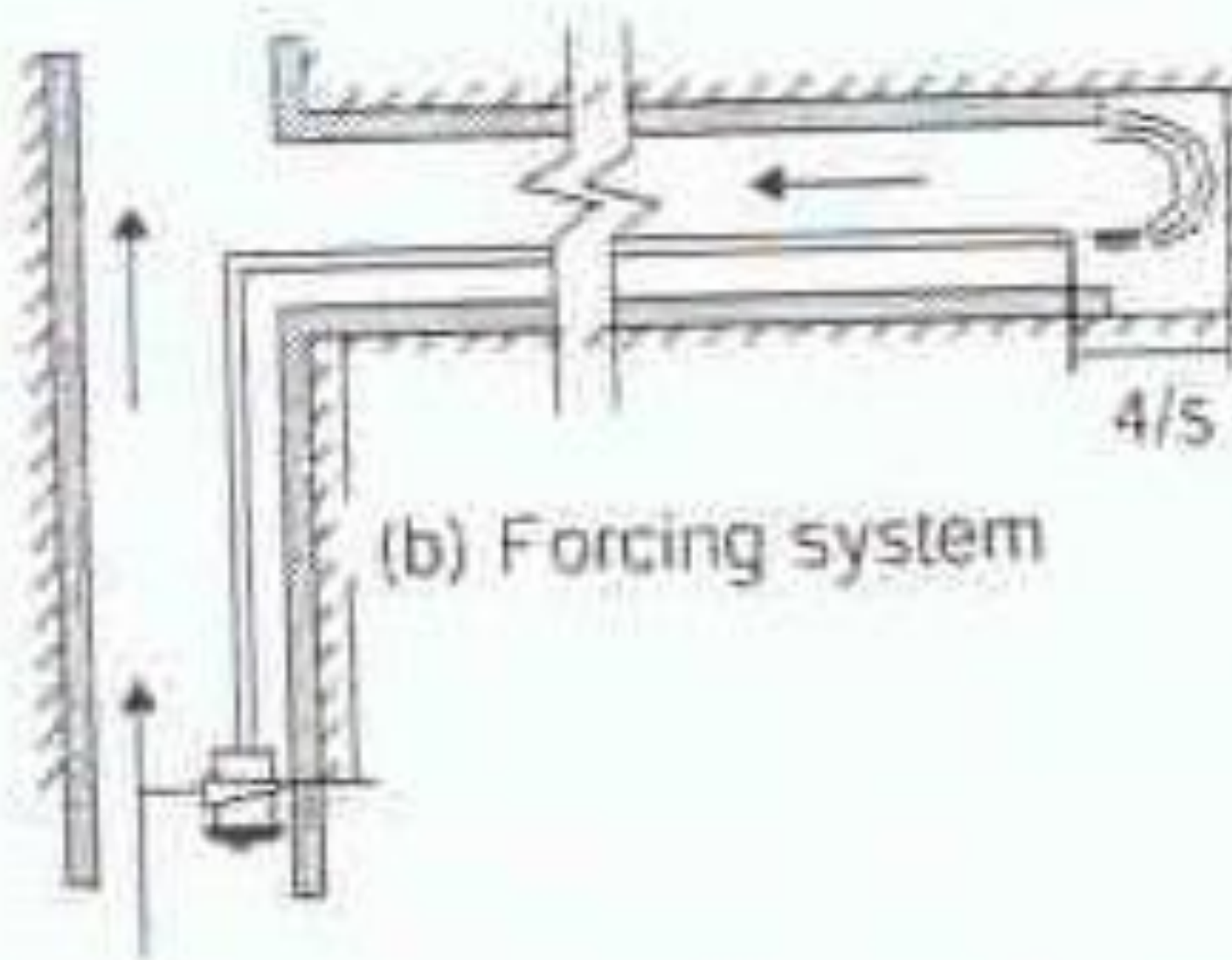
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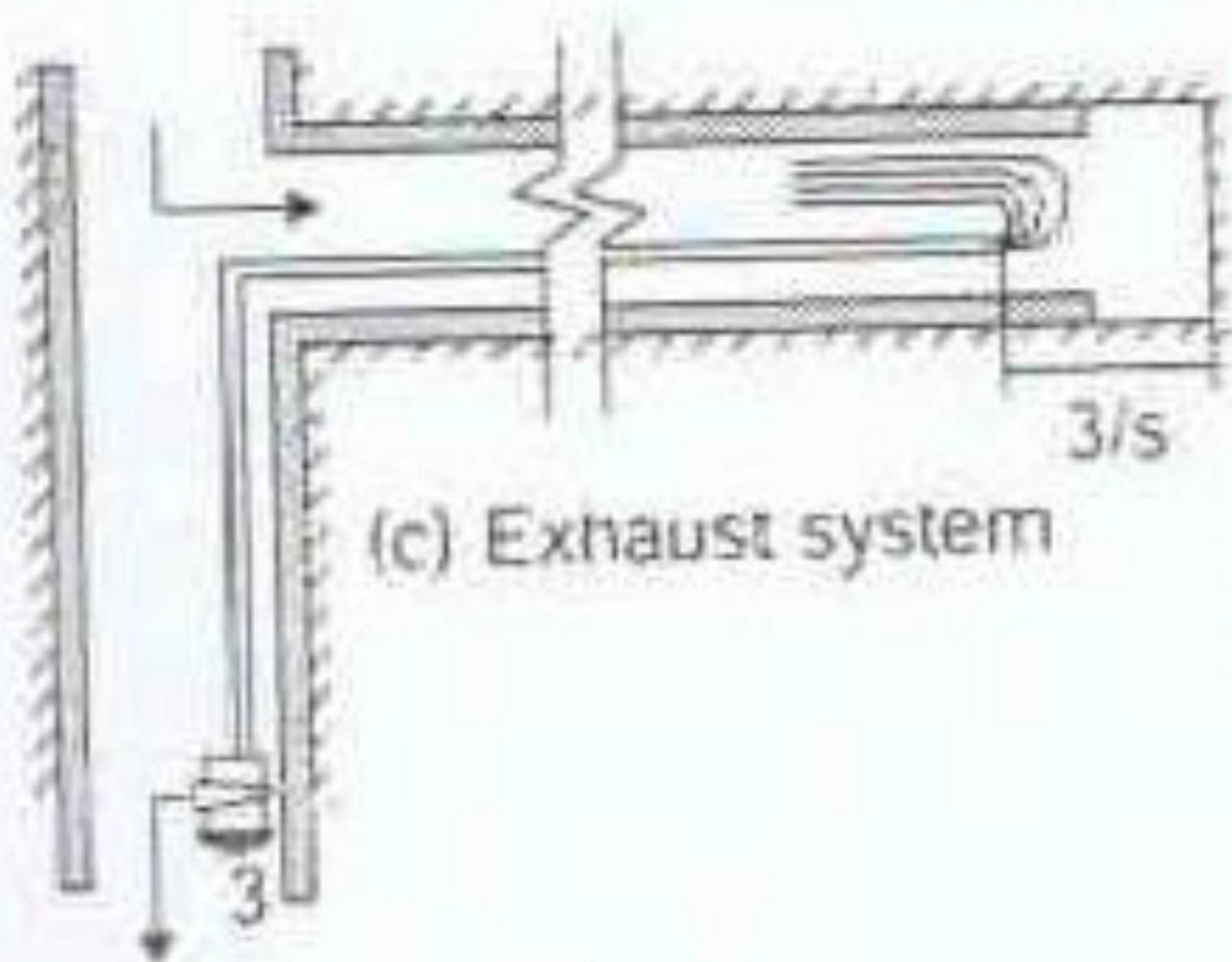
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Ventilation Forcing System

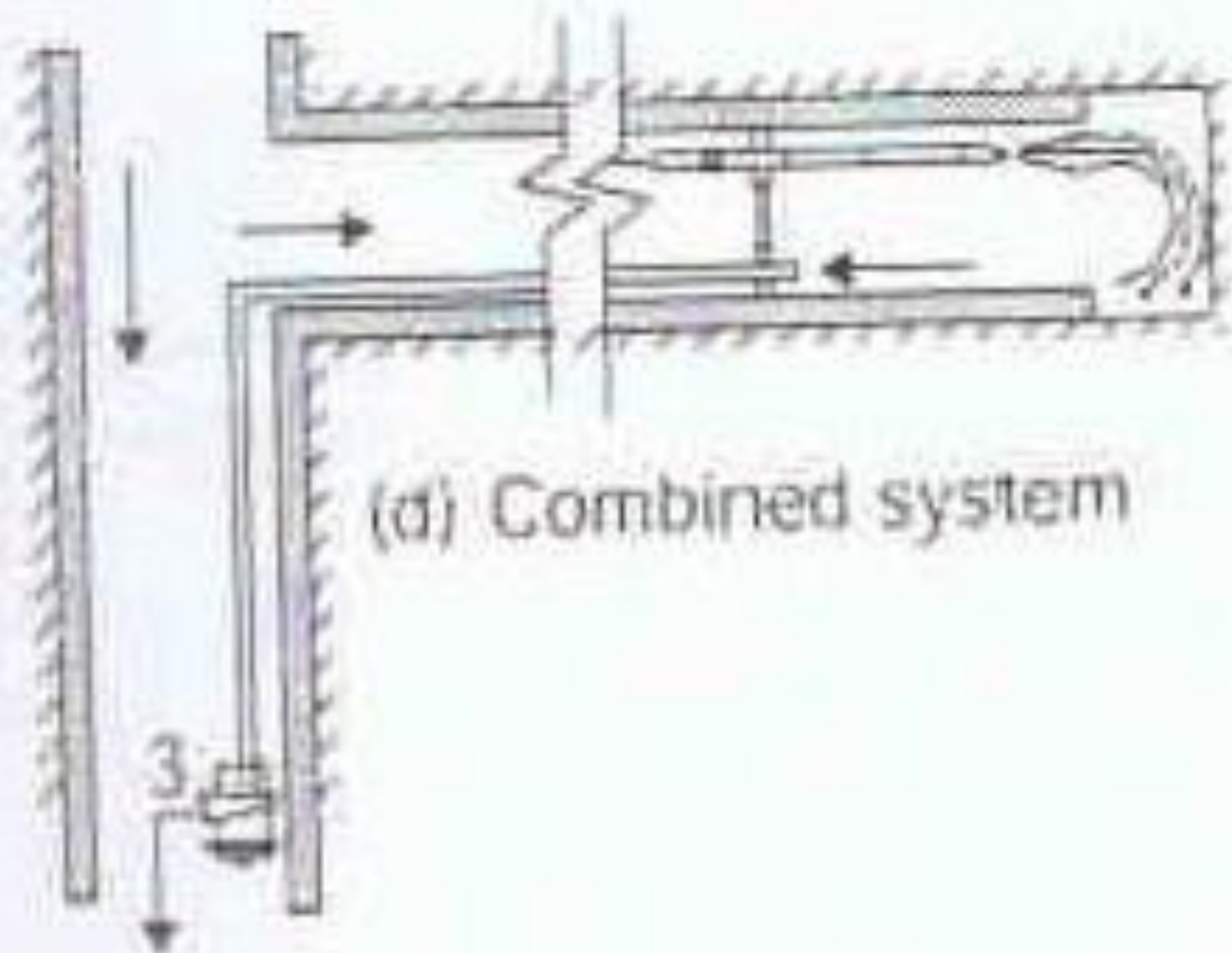


Ventilation Exhaust System



Ventilation Combined System

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(d) Combined system

Auxiliary ventilation in mines.

Temporary Drive Support I

- *While driving in mine, some time situation so arises that with out support even for few minutes to few hour, drive can not stand.
- *In such situation, immediate temporary support is must.
- *This condition becomes very prominent, when excavation is near surface or in underground very weak, unstable formation.
- *This situation demands, advance support before digging ground ahead of drive.
- *Fore poling method is used in this situation.

Temporary Drive Support II

- *Timber pole can be hammered in soft ground / soft rock formation to form an artificial cover while excavating soft formation.
- *now steel pipes are common for fore poling in place of wood / timber.
- *Long holes are drilled all along periphery of drive.
- *Steel pipes are inserted in these holes.
- *Immediately, after removal of soft rock below fore pole, RCC arch is erected as permanent support to drive.

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Temporary Drive Support III

- * In case, where ground condition is not that bad, temporary support like chock mate, crib, timber pole etc are erected before placing permanent support.

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Permanent Drive Support I

- *Permanent support is in use for protection of drive & safety of persons.
- *Some time permanent support is used for drive, when it is mandatory by law.
- *Permanent support can be lining, rock bolts, steel joist, wire mesh, shot crete etc.
- *Often combination of support is used based on requirement.
- *Type & nature of permanent support also depends on importance, purpose & life span of drive.

Natural Permanent Support I

- *Natural support – rock it self act as support & no additional support is required.
- *Rock has to be of good competent rock in low stressed condition with respect to rock strength.

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Rock Reinforcement I

- *Rock bolts, cable bolts, rock stitching are used for rock reinforcement.
- *Use of rock bolts in drive or any excavation in mine is increasing.
- *Rock bolts are used in good quality rock as permanent support.
- *Tension improves effectiveness of rock bolts.
- *In weak ground cement & resign anchored rock bolts are suitable.
- *Mechanical anchors requires competent ground for suitable anchorage.

Rock Reinforcement II

*Rock bolts are also as temporary supports & to be substitutes by additional support in future.

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Steel Joist Support I

- *In main drive, where rock mass is fractured & inherently weak to extend that rock bolts can not function effectively, steel arches are used.
- *Some time steel arches are used with good rock mass at depth.
- *At depth, rock mass exposed to drive are squeezed & gets deformed.
- *In this situation. Yielding steel arches can extend effective support even after yielding.
- *Rolled Steel Joist (RJS) is used for steel arches.
- *Steel – joist, steel arch

Steel Joist Support II

- *Steel arches are made from desired shape & size of RJS.
- *Arches can be fabricated at site or can be supplied by suppliers.
- *“H” section RJS are used for rigid arches.
- “V” sections are used for yielding arches.
- *These arches are placed at distance of 0.5m to 1.25m from each other.
- *Lagging behind arches could be wood, RCC slabs of various shape & sizes, steel members.

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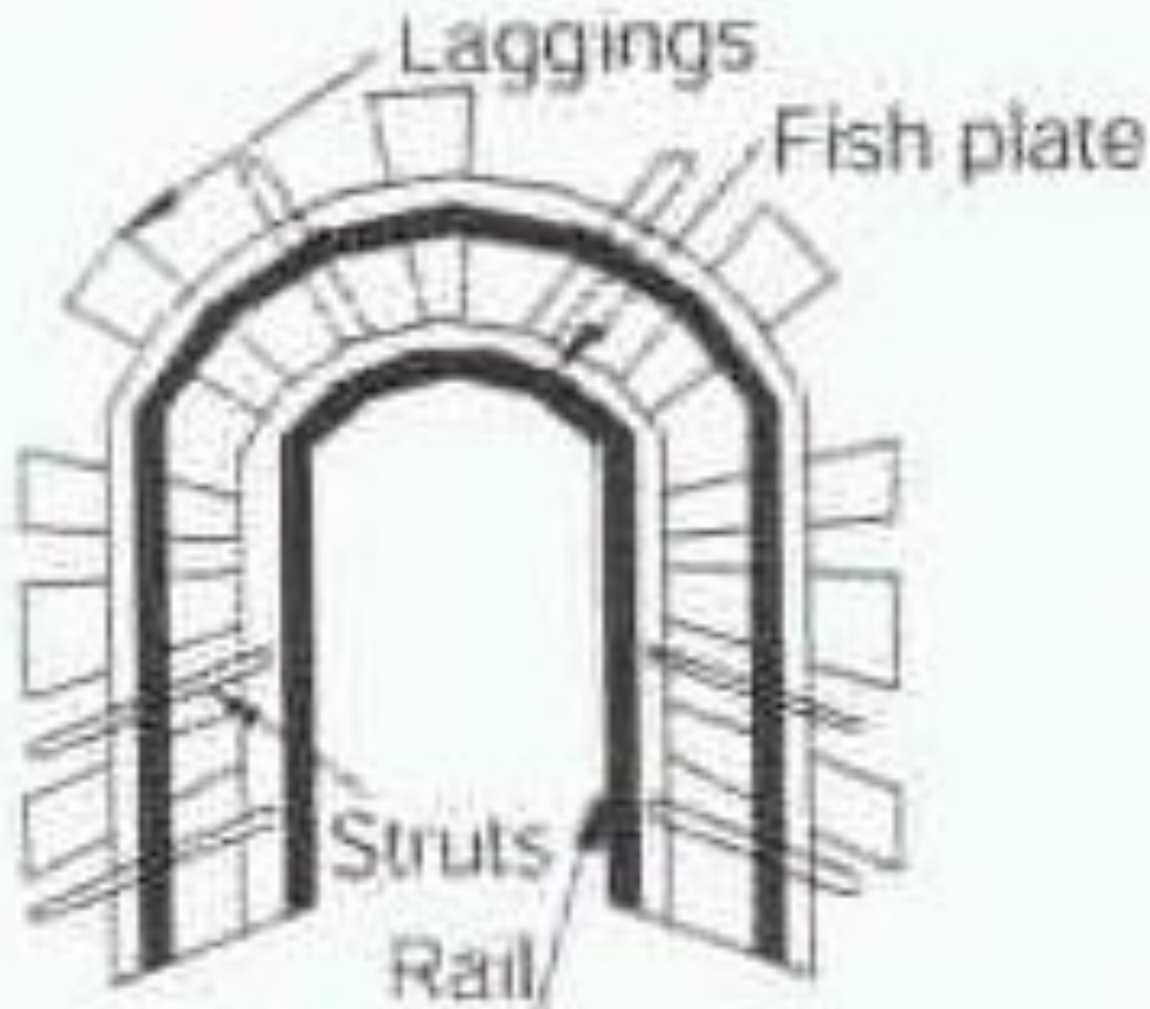
Steel Joist Support III

*Individual arches are tied to each other for better support.

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Steel Arch Rigid I

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(b) Steel arches

Wood Support I

- *Wooden support is mostly used as temporary support.
- *In old days wooden support were used as permanent support.
- *Wooden support – is now becoming out dated.
- *It takes lot of space in excavation resulting restriction in movement & flow of air in drive.
- *Longevity of wood support is very less, specially in humid & wet climate.
- *Wooden support puts restriction in air ventilation flow.

Wood Support II

- *Wooden support is vulnerable to fire at dry condition.
- *Wooden support puts restriction in air ventilation flow.
- *Longevity of wood support is very less, specially in humid & wet climate.
- *Wood gets rotten very fast in hot, humid air.

Concrete Support I

- *Concrete – monolithic (plain) concrete. Reinforced concrete,
- *In suite continuous RCC arche is used in underground drive as permanent support.
- * In suite RCC arche is in very close contact with rock, thus protecting rock from weathering & yielding.
- *RCC arch can yield with pressure but continues to offer support.

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Concrete Support I

*Concrete – monolithic (plain) concrete. Reinforced concrete,

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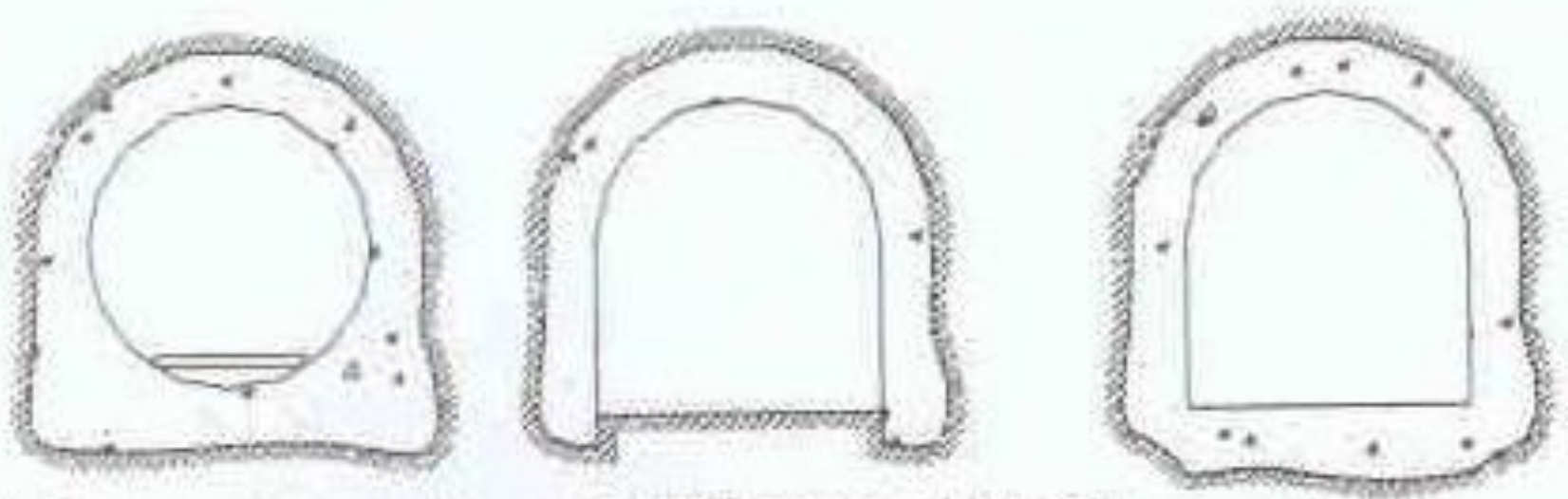
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Concrete Arches

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(c) Concrete arches of different shapes

Shotcrete Permanent Support I

- *Shot crete is used as surface coating to prevent rocks from further deuteriation due to ventilation air current.
- *It is also used as covering of steel , wood support etc.
- *Shot crete is used as logging between excavated rock surface & support that has been erected.
- *Shot crete (secondary lining) does not take any load.
- *This lining is used to provide aesthetic finish & sealent, fire resistant coating or protective covering to temporary & permanent support.

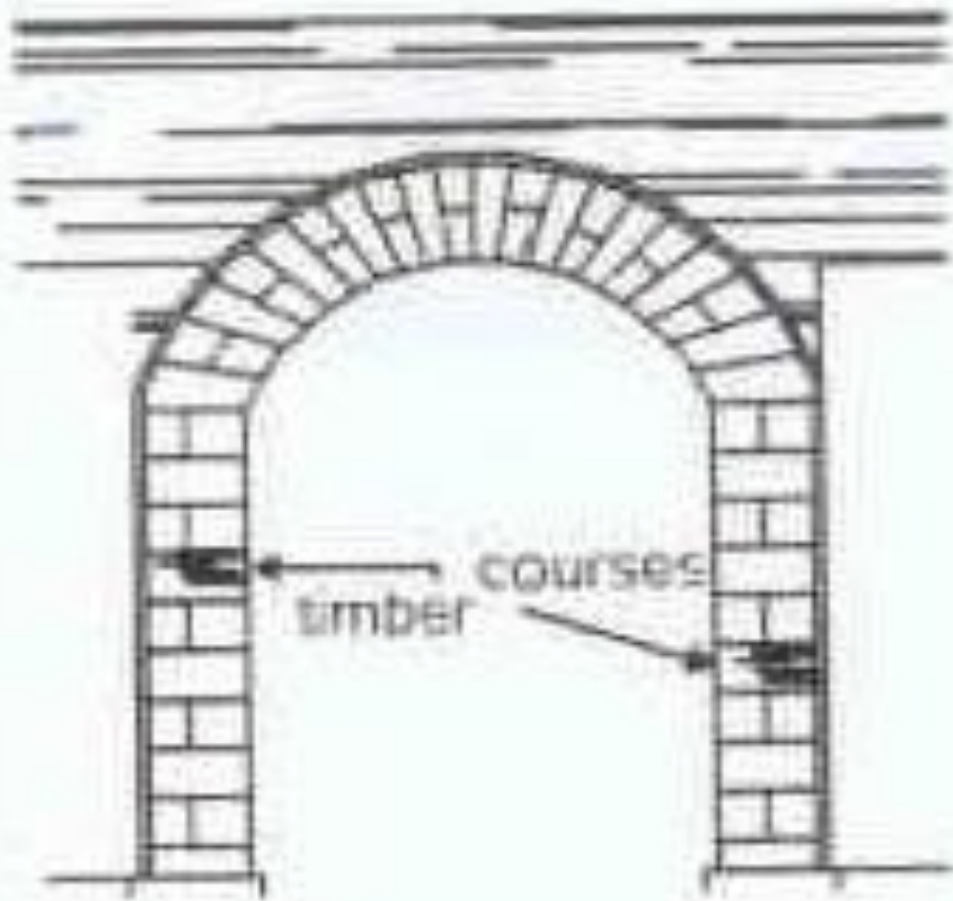
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Segment Support I

*Segment support – tubbing made of cast steel, cast iron, reinforce concrete, brick, pre fabricated concrete,

Yielding Brick Arch

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(a) Brick arch with timber to provide yielding

Reference I

*Surface and Underground Excavation- Methods,
Techniques and Equipment by Ratan Raj Tatiya.
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